

IS THE DIMINISHED ATTENTIONAL BLINK FOR SALIENT T2 STIMULI DRIVEN BY A RESPONSE BIAS?

Helen Tibboel, Jan De Houwer, and Bram Van Bockstaele

Ghent University

Bruno Verschuere

Ghent University, University of Amsterdam, and Maastricht University

The attentional blink (AB) effect refers to the finding that the second of two quickly presented masked targets (T1 and T2) is less likely to be identified when the temporal lag between the targets is short than when it is long (Raymond, Shapiro, & Arnell, 1992). Interestingly, the AB effect is diminished for salient T2 stimuli. The common explanation is that salient stimuli are processed more efficiently than other stimuli. A contrasting hypothesis is that it is due to a bias to report such stimuli. Recently, Tibboel, Van Bockstaele, and De Houwer (2011) introduced a signal detection method that allowed them to reject the response bias explanation for the effect of arousing words on the AB. Presently, we used the same method to examine the diminished AB for participants' own names (Experiment 1) and stimuli that form a coherent category (Experiment 2). Both experiments confirmed that the effect of these stimuli on the AB reflects more efficient processing rather than a response bias.

Key words: attentional blink, rapid serial visual presentation, salience, response bias

In the attentional blink (AB) task, participants are required to identify two masked targets (T1 and T2) that are presented in a rapid serial visual presentation (RSVP) stream. The AB effect refers to the finding that T2 is identified less well when the temporal interval (i.e., lag) between T1 and T2 is short (i.e., between approximately 100 and 400 ms) than when it is long (i.e., 400 ms or more; Raymond, Shapiro, & Arnell, 1992). This effect was first assumed to reflect the structural limitations of the human attentional system (e.g., Isaak, Shapiro, & Martin, 1999; Potter, Staub, & O'Connor, 2002; Raymond et al., 1992). Subsequent research showed that the AB effect is diminished when T2 is a salient stimulus, thus implying that there is no structural limitation. For instance, the AB is attenuated when T2 is highly arousing (e.g., Anderson, 2005; Keil & Ihssen, 2004). Furthermore, the AB is smaller when T2 is highly familiar, like the participant's own name (Shapiro, Caldwell, & Sorensen, 1997), a familiar face (Jackson & Raymond, 2006), or an item for which the

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Correspondence concerning this article should be addressed to Helen Tibboel, Department of Psychology, Ghent University, Henri Dunantlaan 2, B-9000 Ghent, Belgium; E-mail: Helen.Tibboel@UGent.be

identification was practiced intensively (Maki & Padmanabhan, 1994). Finally, Tibboel, De Houwer, Spruyt, and Crombez (2011) recently argued that T2 stimuli do not have to be intrinsically salient to be less affected by the AB. In addition, stimuli that are salient because of the context in which they are presented result in a smaller AB effect. More specifically, Tibboel et al. showed that the AB was smaller for T2 stimuli that were neither affective nor extremely familiar, but that did form a coherent category within the T2 stimulus set (e.g., when half of the T2 stimuli are number words and the other T2 stimuli share no obvious feature).

The effects of salient stimuli on the AB are typically explained in terms of more efficient processing of salient stimuli than non-salient stimuli (e.g., Anderson, 2005; Jackson & Raymond, 2006; Shapiro et al., 1997; Tibboel, De Houwer, et al., 2011). However, it is possible that the modulation of the AB by salient stimuli is due to a response bias. Because participants often do not know the identity of T2 when it is presented at short lags after T1, it is possible that they adopt a guessing strategy to deal with this situation. When a specific word (e.g., the name of the participant) or specific categories of words (e.g., arousing taboo words or number words) occur frequently in an experiment, participants might be inclined to report those words whenever they are unsure about the identity of T2. Such a response bias would increase the number of correct responses on trials with a salient stimulus, especially on short lags where the probability of a correct identification of T2 is small. As such, the reduced AB for salient stimuli might be due to a response bias rather than to the more efficient processing of those stimuli.

Tibboel, Van Bockstaele, and De Houwer (2011) recently examined the response bias explanation for the modulation of the AB by arousing taboo T2 stimuli. Rather than asking participants to type in T2, Tibboel et al. showed participants an array of words and asked them to select from that array the word that corresponded to T2. This procedure allowed the authors to perform signal detection analyses on the data (Wickens, 2002). First, they calculated an index of the sensitivity for the taboo and neutral T2 stimuli (d'). Second, they calculated an index of the bias to report a taboo or neutral stimulus ($\log \beta$). Their analyses showed that d' was higher for taboo T2 stimuli than for neutral T2 stimuli and that this difference was larger at the short lag. Thus, Tibboel et al.'s results clearly implied that the reduced AB effect for arousing taboo words was due to a more efficient processing of taboo stimuli than neutral stimuli. The analyses of $\log \beta$ revealed that there was no bias to report taboo T2 stimuli. In fact, there was a bias *against* reporting taboo stimuli. Controlling for this bias thus revealed an even stronger modulation of the AB by taboo T2 stimuli, which suggests that this modulation was underestimated in previous emotional AB studies. Possibly the bias against reporting taboo words in the experiment of Tibboel et al. was due to participants' concern that reporting highly sexual taboo words would be considered inappropriate. This study has thus proven that the signal detection method is a valuable method for analyzing AB data. It is crucial that researchers who use the AB paradigm to study the special attentional status of specific salient stimuli examine the possibility that effects are partly due to a bias to report those stimuli. Similarly, possible null findings could be due to a bias not to report such stimuli.

Even though Tibboel, Van Bockstaele, and De Houwer (2011) have shown that the diminished AB for taboo T2 stimuli cannot be attributed to a bias to report such stimuli, the question of whether a response bias affects the modulation of the AB by salient stimuli in general remains far from resolved. Because other salient stimuli such as participants' own names (Shapiro et al., 1997) and words that form a coherent category (Tibboel, De Houwer, et al., 2011) are very different from taboo words (e.g., they are not arousing, and it is not socially unacceptable to utter these words), it is necessary to perform separate experiments to examine response bias for these stimuli. We therefore conducted two experiments in which we examined whether the attenuation of the AB for participants' own names (Experiment 1) and for words that form a coherent category (Experiment 2) is due to a response bias. Signal detection analyses allowed us to distinguish between sensitivity and response bias and thus to establish whether these two types of stimuli are

truly processed more efficiently than control stimuli. Our studies are important not only to clarify the reasons why AB effects are moderated by the nature of the stimuli but also to shed new light on the attentional status of own names and stimuli that form coherent categories. Definite conclusions about the latter issue cannot be reached on the basis of existing studies.

Experiment 1

Previous AB studies do not provide conclusive evidence for the special attentional status of own names because the reduced AB for those stimuli could have been driven by a response bias. We therefore designed an experiment that was highly similar to Experiment 2 of Shapiro et al. (1997), but performed signal detection analyses. By controlling for a response bias, our first experiment provides a new and more stringent test of the hypothesis that own names have a special attentional status.

Method

Participants. Twenty-two psychology students at Ghent University (four men) participated in this experiment in exchange for course credits. Only participants whose names were four ($N = 15$) or five ($N = 7$) letters long were invited to participate. The mean age was 18.14 years ($SD = 0.71$, range 17–21). All participants gave their informed consent.

Apparatus. The AB task consists of distractors and T1 and T2 stimuli. All stimuli were common boys' and girls' names that consisted of four or five letters. Just like in the study by Shapiro et al. (1997), 30 distractors (15 girls' names, 15 boys' names; 15 four-letter names, 15 five-letter names) and 10 T1 stimuli (5 girls' names, 5 boys' names; 5 four-letter names, 5 five-letter names) were presented. T2 was the participant's own name in one block and another participant's name in another block.¹ Unlike Shapiro et al., we asked participants to write down the names of their parents, grandparents, siblings, partner(s), ex-partner(s), in-laws, best friends, neighbors, pets, and otherwise important persons in their lives before the experiment started. This allowed the experimenter to check whether any of the names also occurred on the list of T1 stimuli or distractor stimuli. If this was the case, we replaced the name on these lists with another name that was matched for length and gender. All names that were used in the experiment can be found in Appendix A. All stimuli were presented in upper-case letters in 7-point Courier New font on a 17-in, 85-Hz monitor. For stimulus presentation and response registration, we used the E-Prime software package (Schneider, Eschman, & Zuccolotto, 2002a, 2002b).

Procedure. Participants were tested in rooms that contained three or four separate cubicles in which computers were set up. One, two, three, or four participant(s) were tested during each session. They were seated at a distance of approximately 50 cm from the computer screen. Each trial started with the presentation of the sentence "Press the spacebar to start the trial." After pressing the spacebar, participants saw a red fixation cross in the center of the screen for 1,000 ms. The fixation cross was followed by the RSVP stream, consisting of 15 stimuli. T1 was always presented in white. Distractors and T2 were always black. On half of the trials (i.e., the "T2 present" trials), 13 distractor names and the T2 name (participants' own names in the "own name" block, another participant's name in the "control name" block) were presented. In the other trials ("T2 absent" trials), there were 14 distractors and no T2. Each stimulus in the sequence was presented for 59 ms (equaling six screen refreshes), with a 12-ms interstimulus interval. All stimuli appeared in the center of the screen, against a grey background.

1 Two male participants who participated during the same session were supposed to be presented with each other's names as a control name. However, because they were friends, and friends' names are likely to stand out, a control name that did not belong to another participant was used.

Participants were instructed to monitor the stream. They were asked to report the white word and to indicate whether the T2 name was present or absent. At the end of each trial, the question “What was the white word?” appeared. Participants were required to type in the name that they thought they had seen. After this, the question “Was [name] present or not present?” appeared. This time, participants were required to press “c” when they thought the name was present and “n” when they thought it was not. The experiment continued after the participants answered both questions. There was no response deadline.

T1 and the distractors were randomly selected from the list of T1 stimuli and the list of distractor stimuli, respectively. The order of trials was randomized. T1 could appear at the third, fourth, or fifth position in the stream, and T2 could appear two or eight lags after T1 (i.e., with one or seven intervening distractors), reflecting a stimulus onset asynchrony (SOA) of 142 and 568 ms, respectively. There were two separate blocks, the order of which was counterbalanced across participants. In one block, T2 was the participant’s own name. In the other block, another participant’s name was presented. The two names were always of the same gender and contained an equal number of letters. In each block, each of the 12 trial types (T2 present or absent * Lag 2 or Lag 8 * T1 at position 3, 4, or 5) was repeated four times, yielding 48 trials per block—a total of 96 experimental trials. At the beginning of each block, there was a practice block consisting of 12 trials.

Data analysis. We calculated the proportion of correct responses for both experimental conditions (own name, control name). For the analysis of T2 responses, only trials with a correct identification of T1 were taken into account. The proportion of hits was calculated by dividing the number of times the participant correctly indicated that the name was present by the number of times this name was present. The proportion of false alarms was calculated by dividing the number of times that a participant incorrectly indicated that the name was present by the number of times that the name was absent.

As in the study by Tibboel, Van Bockstaele, and De Houwer (2011), scores of 0 were replaced by 0.001 and scores of 1 were replaced by 0.999.² We then calculated d' and $\log \beta$ for both participants’ own names and the control name, using the following formulas:

$$d' = Z(h) - Z(f) \text{ and} \\ \log \beta = \frac{1}{2}[Z^2(f) - Z^2(h)],$$

where “h” stands for *hits* and “f” for *false alarms*. The more positive the value of d' , the more sensitive the participant is to process the signal. Positive $\log \beta$ scores indicate a bias to say that the signal was absent; negative $\log \beta$ scores reflect a bias to say that the signal was present (Wickens, 2002).

We performed four separate repeated-measures analyses of variance (ANOVAs; on hits, false alarms, d' , and $\log \beta$), with T2 type (own name or control name) and lag (2 or 8) as within-subjects factors.

Results

We excluded the data of one participant because the proportion of correct T1 identifications ($M = .13$) was more than 2.5 standard deviations below the group mean ($M = .85$, $SD = .19$). The identification of T1 did not differ between conditions, all F s < 1. Means can be found in Table 1.

² In Experiment 1, several cells in our data matrix contained a value of 0 (i.e., there were no hits/false alarms at all) or 1 (i.e., there were only hits/false alarms). We replaced these values with 0.001 and 0.999, respectively, solely for the purpose of normalization. Because we replaced these values for both conditions, the crucial interactions cannot be affected by this procedure. The following changes were made in the analysis of the hits: In the control block, 19% of the cells contained a proportion of 0 hits on Lag 2 and 5% contained a proportion of 0 hits on Lag 8. Ten percent of the cells on Lag 8 had a value of 1. In the own name block, 38% of the cells at Lag 2 and 43% of the cells at Lag 8 contained a value of 1, whereas there were no values of 0. In our analyses of the false alarms, the following values were replaced: In the control condition, 52% of the cells at Lag 2 and 43% of the cells at Lag 8 contained a value of 0. In the own name condition, 62% of the cells at Lag 2 and 57% of the cells at Lag 8 contained a value of 0. Note that when we replaced the values with 0.01 and 0.99, results were virtually the same.

Table 1
Proportion of Hits, False Alarms, d' , and Log β for T2 Given Correct Identification of T1 for Experiment 1

T2 type	Lag 2		Lag 8	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Hits				
Other name	0.22	0.20	0.67	0.28
Own name	0.85	0.17	0.88	0.21
False alarms				
Other name	0.06	0.07	0.10	0.14
Own name	0.06	0.09	0.09	0.13
d'				
Other name	1.04	1.33	2.49	1.69
Own name	3.96	1.69	3.98	1.96
Log β				
Other name	1.62	2.46	1.43	2.75
Own name	1.10	2.97	0.46	2.72

Hits. Most important, the interaction between T2 type and lag was highly significant, $F(1, 20) = 38.11$, $p < .001$, $\eta_p^2 = .81$. Participants performed significantly worse at Lag 2 than at Lag 8 for the control name (i.e., AB), $t(20) = 7.27$, $p < .001$, whereas there was no difference between performance on Lag 2 and Lag 8 for own name (i.e., there was no AB effect), $t < 1$. A main effect for T2 type, $F(1, 20) = 85.28$, $p < .001$, $\eta_p^2 = .67$, revealed that performance in the own name condition was overall better than in the control name condition. Finally, there was also a main effect for lag, $F(1, 20) = 29.67$, $p < .001$, $\eta_p^2 = .60$, showing better performance at Lag 8 than at Lag 2.

False alarms. There was no interaction between condition and lag, nor was there a main effect for T2 type, $F_s < 1$. The main effect for lag approached significance, $F(1, 20) = 3.51$, $p = .08$, $\eta_p^2 = .15$, suggesting there were slightly more false alarms at Lag 8.

d' . The results of our d' analyses mirrored those of the hits. There was an interaction between T2 type and lag, $F(1, 20) = 7.63$, $p < .05$, $\eta_p^2 = .28$. There was no lag effect in the own name block, $t < 1$, whereas performance at Lag 2 was significantly worse than performance at Lag 8 in the control name block, $t(20) = 3.65$, $p < .005$. A main effect of T2 type, $F(1, 20) = 31.72$, $p < .001$, $\eta_p^2 = .61$, showed that d' was larger in the own name condition than in the control name condition. Finally, the main effect of lag, $F(1, 20) = 7.20$, $p < .05$, $\eta_p^2 = .27$, showed that d' increased with lag.

Log β . Our analyses of log β did not reveal any significant effects, $F_s < 2.51$.

Discussion

In line with Shapiro et al. (1997), we found no AB effect for participants' own names, whereas we did find a strong AB effect for control names. The finding that there was no AB effect at all for participants' own names illustrates the special attentional status of this stimulus. Most important, we were able to confirm this with our signal detection method. Both the analyses of the hits and the analyses of d' showed that the modulation of the AB by own names is not due to a response bias. Our analyses of false alarms and log β even show that there was no difference in participants' bias to report their own name compared to another participant's name. The effect size for the crucial interaction between condition and lag was numerically smaller for the analyses of d' than for the analyses of the hits, suggesting that d' is a more conservative measure.

Experiment 2

Although Tibboel, De Houwer, et al. (2011) demonstrated for the first time that the AB is reduced for T2 stimuli that form a coherent category, they could not exclude the possibility that their effects were partly due to a response bias. For instance, their second experiment revealed a smaller AB effect for words referring to numbers than for random words. The analyses of participants' incorrect responses, however, suggested that participants more often reported words referring to numbers when they were guessing. Hence, the reduced AB for numbers could have been due to a response bias. We therefore replicated the study of Tibboel et al. in a way that allowed for signal detection analyses.

Method

Participants. Twenty-three psychology students at Ghent University (one man) participated in this experiment in exchange for course credits. The mean age was 19.35 years ($SD = 2.60$, range 17–30). All participants gave their informed consent.

Apparatus. There were six different T2 words referring to numbers and six different random T2 words (see Appendix B). The two types of words did not differ in length ($M = 4.50$, $SD = 0.84$, and $M = 4.83$, $SD = 0.75$, respectively, $t < 1$), nor did they differ in frequency per million ($M = 211.17$, $SD = 157.16$, and $M = 201.50$, $SD = 106.11$, respectively, $t < 1$). There were 30 neutral T1 words (word length: $M = 5.70$, $SD = 1.18$) and 79 neutral distractors (word length: $M = 12.73$, $SD = 2.07$). The distractors were adapted from stimuli used by Anderson (2005). We used relatively long distractors in order to sufficiently mask the targets. All words were Dutch. All stimuli were presented in capital letters in 16-point Courier New font on a 17-in, 85-Hz monitor. T1 and T2 were presented in green, the distractors were white, and the background was black.

Procedure. Whereas the procedure of Experiment 1 was based on the experiments of Shapiro et al. (1997), the procedure of Experiment 2 was highly similar to Tibboel, De Houwer, et al. (2011). This is the reason there are important differences between the procedures of Experiment 1 and Experiment 2. First, in Experiment 2, a T2 stimulus was always present and participants were required to identify both T1 and T2. Second, each stimulus in the sequence was presented for 94 ms (equaling eight screen refreshes) in the center of the screen. Third, at the end of each trial, the question "What was the first green word?" appeared. Underneath it, all 30 T1 stimuli were presented in 18-point bold Courier New font, arranged in three columns and ordered alphabetically. Participants were asked to click the word that they thought was presented. After this, the question "What was the second green word?" appeared. This time, all 12 T2 targets were presented on the screen, in 18-point bold Courier New font, arranged in three columns and ordered alphabetically. Again, participants were asked to click the word that they thought they had seen. The experiment continued after the participant clicked the second word. There was no response deadline.

T1 and the distractors were randomly selected from the list of T1 stimuli and the list of distractor stimuli, respectively. On half of the trials, T2 was a word referring to a number, and on the other half of the trials, T2 was a random word. T1 could appear at the third or fifth position in the stream, and T2 could appear two or eight lags after T1 (i.e., with one or seven intervening distractors), reflecting an SOA of 188 ms and 752 ms, respectively. There were three blocks in which each of the 12 T2 stimuli was presented once at each lag, once when T1 was presented on the third position, and once when T1 was presented on the fifth position. This yielded 48 trials per block, resulting in 144 experimental trials in total. The order of the trials was randomized. There was also a practice block that consisted of four trials in which neutral words were presented.

Data analysis. The data analyses were similar to Experiment 1, with the following exceptions. We calculated the proportion of correct responses for all stimuli separately. The proportion of hits was calculated by dividing the number of times the participant correctly indicated that a specific stimulus was present by the number of times this

stimulus was presented. The proportion of false alarms was calculated by dividing the number of times that a participant incorrectly indicated that a specific stimulus was presented by the number of times that the stimulus was not presented. For both types of T2 stimuli (numbers and random), we calculated the average number of hits and false alarms. Next, we calculated d' and $\log \beta$ for all stimuli separately before averaging these indices for both T2 types. Again, values of 0 were replaced by .001, and values of 1 were replaced by .999.³ We performed four separate repeated-measures ANOVAs (on hits, false alarms, d' , and $\log \beta$) with T2 type (number or random) and lag (2 or 8) as within-subjects factors.

Results

We excluded the data of one participant because the proportion of correct T1 identifications ($M = .69$) was more than 2.5 standard deviations below the group mean ($M = .91$, $SD = .07$). A repeated-measures ANOVA on the T1 data with T2 type and lag as within-subjects factors revealed that the identification of T1 tended to be affected by lag, $F(1, 21) = 4.04$, $p = .06$, $f^2 = .19$, showing slightly better performance at Lag 8, $M = .93$, $SD = .05$, compared to Lag 2, $M = .91$, $SD = .05$, other F s < 1 . Relevant means of the analyses of T2 data can be found in Table 2.

Table 2
Proportion of Hits, False Alarms, d' , and Log β for T2 Given Correct Identification of T1 for Experiment 2

T2 type	Lag 2		Lag 8	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Hits				
Control	0.43	0.24	0.94	0.05
Coherent	0.59	0.23	0.95	0.05
False alarms				
Control	0.04	0.02	0.01	0.01
Coherent	0.05	0.02	0.01	0.00
d'				
Control	1.65	1.46	5.36	0.59
Coherent	2.30	1.36	5.39	0.54
Log β				
Control	1.04	1.31	0.55	0.73
Coherent	0.73	0.79	0.36	0.70

Hits. Importantly, the interaction between T2 type and lag was significant, $F(1, 21) = 26.86$, $p < .001$, $\eta_p^2 = .56$. Performance for T2 number words was better than for random words at Lag 2, $t(21) = 4.89$, $p < .001$, but not at Lag 8, $t < 1$, indicating a reduced AB effect for number words compared to random words. For both the T2 number words and the random words, performance was better at Lag 8, $t(21) = 7.86$, $p < .001$, than at

3 In Experiment 2, again, several cells in our data contained a value of 0 (i.e., there were no hits/false alarms at all) or 1 (i.e., there were only hits/false alarms). The following changes were made in the analysis of the hits: For the coherent category, 4% of the cells contained a proportion of 0 hits on Lag 2 and 0% contained a proportion of 0 hits on Lag 8. 18 percent of the cells on Lag 2 and 75% of the cells on Lag 8 contained a value of 1. For the random words, 14% of the cells at Lag 2 and 0% of the cells at Lag 8 contained a value of 0, and 6% of the cells at Lag 2 and 72% of the cells at Lag 8 contained a value of 1. In our analyses of the false alarms, the following data were replaced: For the words that formed a coherent category, 12% of the cells at Lag 2 and 72% of the cells at Lag 8 contained a value of 0. For the random words, 20% of the cells at Lag 2 and 78% of the cells at Lag 8 contained a value of 0. None of the cells contained a value of 1.

Lag 2, $t(21) = 11.14$, $p < .001$. There was also a main effect for lag, $F(1, 21) = 101.10$, $p < .001$, $\eta_p^2 = .83$, indicating that performance was better at Lag 8 than at Lag 2, and a main effect for T2 type, $F(1, 21) = 17.75$, $p < .001$, $\eta_p^2 = .46$, showing that performance was better for words referring to numbers than for random words.

False alarms. Again, there was a significant interaction between T2 type and lag, $F(1, 21) = 8.48$, $p < .01$, $\eta_p^2 = .29$. There were significantly more false alarms for words referring to numbers than for random words at Lag 2, $t(21) = 2.65$, $p < .05$, but not at Lag 8, $t < 1$, suggesting that there is a possibility that the hit rate differences are due to a bias to report more number words. Furthermore, there was a significant lag effect, $F(1, 21) = 69.92$, $p < .001$, $\eta_p^2 = .82$, showing that there were more false alarms at Lag 2 than at Lag 8. A significant main effect for T2 type, $F(1, 21) = 4.63$, $p < .05$, $\eta_p^2 = .18$, revealed that there were more false alarms for words referring to numbers than for random neutral words.

d' . The outcome of our analyses of d' mirrored those of the analyses of the hits. There was a significant interaction between T2 type and lag, $F(1, 21) = 19.04$, $p < .001$, $\eta_p^2 = .48$. d' for words referring to numbers was larger than d' for random words at Lag 2, $t(21) = 4.47$, $p < .001$, but not at Lag 8, $t < 1$. There was also a significant lag effect, $F(1, 21) = 169.55$, $p < .001$, $\eta_p^2 = .89$, revealing that d' was larger as lag increased. The main effect for T2 type was also significant, $F(1, 21) = 10.12$, $p < .005$, $\eta_p^2 = .33$. Performance was better for words referring to numbers than for random words.

Log β . T tests reveal that both for words referring to numbers at Lag 2 and at Lag 8 and for random words at Lag 2 and at Lag 8, there was a bias to choose a T2 other than the one that was actually presented, $t(21) = 4.29$, $p < .001$, $t(21) = 2.38$, $p < .05$, $t(21) = 3.72$, $p < .005$, and $t(21) = 3.57$, $p < .005$, respectively. Importantly, the analyses of log β revealed no significant interaction between T2 type and lag, $F < 1$. There was a marginally significant main effect for lag, $F(1, 21) = 3.09$, $p = .09$, $\eta_p^2 = .13$, showing that the bias to say that another T2 stimulus was presented was stronger at Lag 2 than at Lag 8. However, the lack of an interaction between T2 type and lag, $F < 1$, shows that this cannot explain the modulation of the AB by coherence. Other effects were not significant, $F_s < 2.59$.

Discussion

Our data partly suggest that participants were biased to report words referring to numbers instead of random words. More specifically, there were more false alarms for words referring to numbers at Lag 2. This resulted in a numerically smaller effect size of the crucial interaction between T2 type and lag in the analyses of d' compared to the analyses of the hits. However, the interaction was still highly significant, and the effect size was still large according to the norms of Cohen (1992).

Furthermore, analyses of log β scores revealed no significant effects, confirming that a response bias cannot explain the diminished AB for words that form a coherent category.

General Discussion

The results of our experiments show that the diminished AB for participants' own names and for words that form a coherent category can be found even when controlling for a possible response bias. This has several practical and theoretical implications. First, because we replicated the findings of Shapiro et al. (1997) and the results of Tibboel, De Houwer, et al. (2011), our studies provide additional evidence for the claim that the AB is diminished for these types of stimuli. This new evidence is stronger than previous evidence because our signal detection analyses confirmed that the modulation of the AB by these kinds of stimuli is, at least in part, a genuine attentional effect. Our results thus support the hypothesis of Shapiro et al. and others (e.g., Jackson & Raymond, 2006), who suggest that own names have a low threshold for access to conscious awareness. Furthermore, our research provides us with additional evidence for the conclusion that words that form a coherent category are processed more efficiently than random words (Tibboel, De Houwer, et al., 2011).

Second, even though the results of our analyses of hits and d' did not lead to different conclusions, effect sizes for the d' measure were numerically smaller than effect sizes for the hits. Because d' takes the number of false alarms into account, it provides us with a more accurate index of how efficiently specific salient stimuli are processed. Depending on the nature of the response bias, analyses of d' can reveal stronger (as for the data of Tibboel, Van Bockstaele, & De Houwer, 2011) or weaker effects (as for the current data) than standard analyses. When researchers use the AB task as a tool to measure the extent to which certain stimuli are processed more efficiently than other stimuli in general (e.g., Anderson, 2005; Jackson & Raymond, 2006; Keil & Ihssen, 2004) or in subclinical contexts (Fox, Russo, & Georgiou, 2005; Liu, Li, Sun, & Ma, 2008; Tibboel, De Houwer, & Field, 2010; Trippe, Hewig, Heydel, Hecht, & Miltner, 2007; Waters, Heishman, Lerman, & Pickworth, 2007), we recommend using a signal detection analysis because this allows for a more accurate estimate of the efficiency of stimulus processing.

Third, because our paradigm provides us with a more stringent way to examine whether the AB is modulated by specific stimuli, it is a valuable tool not only for researchers who are interested in practical applications of the AB but also for researchers who are interested in improving theories that are designed to explain the AB. For instance, several models of the AB assume that T2 is processed to a postperceptual level but that it subsequently fails to enter conscious awareness. Some evidence for this hypothesis has been provided by studies using event-related potentials (e.g., Luck, Vogel, & Shapiro, 1996). Importantly, the reduced AB for own names (Shapiro et al., 1997) has also been interpreted as evidence for a postperceptual locus of the AB because semantic processing is necessary in order to observe a different AB effect for one's own name than for other names (see also Giesbrecht, Sy, & Lewis, 2009). Because our results confirm that the AB is modulated by the meaning of the words (i.e., own names vs. other names; words from coherent categories vs. random words), they provide further evidence for a postperceptual processing hypothesis. However, we must note that even though the effects in our Experiment 1 were highly significant, the modulation of the AB by personal names is dependent on the context. For instance, Shapiro et al. (1997) found an attenuated AB for participants' own names in an experiment in which T1 and the distractors were names as well. However, in an experiment in which the distractors and T1 were nouns, they found that the AB was diminished for both participants' own names as well as for other names. Other research (Giesbrecht et al., 2009) has shown that the AB is smaller for participants' own names when the T1 load is low, but not when the T1 load is high. This suggests that the attentional system is flexible in the way it allocates attentional resources to different stimuli. Selection does not always occur at postperceptual processing stages but sometimes at earlier stages as well.

There are two important limitations to our studies. Whereas the procedure of Experiment 1 was highly similar to the original name experiments of Shapiro et al. (1997), we must note that there is an unavoidable difference between the paradigm that we used in Experiment 2 and the paradigms that were previously used (e.g., Anderson, 2005; Fox, Russo, & Georgiou, 2005; Keil & Ihssen, 2004; Liu, Li, Sun, & Ma, 2008; Tibboel et al., 2010; Trippe et al., 2007; Waters et al., 2007). In Experiment 2, all possible T2 targets were presented on a screen, and participants were asked to click the word they thought they had seen. Such a design was necessary in order to perform signal detection analyses. However, in many previous experiments (e.g., Tibboel, De Houwer, et al., 2011), the T2 task was a recall task. Participants were asked to remember the T2 target and to type it in at the end of the trial. In those experiments, participants were not provided with any cues that could help them, whereas in our Experiment 2, they were. Because participants' recognition memory is usually superior to their recall (e.g., Haist, Shimamura, & Squire, 1992), this could have affected our results. Overall performance would thus be better in our task than in the AB tasks in which participants were required to recall T2. However, as there is no reason to assume that this improvement would affect the crucial interaction between T2 type and lag, this does not invalidate our research. Second, the fact that participants were

shown all possible T2 stimuli on the screen at the end of each trial could induce a response bias by making the salient coherent category stand out more than in an experiment in which T2 needs to be recalled. However, if this was indeed the case, this further strengthens the conclusion that the diminished AB for words that form a coherent category is not due to a bias to report these words: Even in a situation where there may be a larger chance that participants become aware of the fact that a specific salient category of stimuli is presented, a response bias cannot explain the modulation of the AB by these salient category exemplars.

To conclude, our research has provided further evidence that salient stimuli such as participants' own names and stimuli that form a coherent category are processed more efficiently than random neutral stimuli. We found that the AB is diminished for salient stimuli and have excluded the possibility that this effect is due to a response bias. We emphasize the benefits of using signal detection analyses of AB data because they provide us with a more accurate way to estimate the extent to which salient stimuli are processed more efficiently than other stimuli. As such, signal detection analyses allow us to draw more solid conclusions and thus have both an important practical and theoretical value.

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Appendix A

Stimuli Experiment 1

T1 names (male):

BART, DAVID, FRANK, KLAAS, KOEN

T1 names (female):

GRIET, INGE, LAURA, LIES, NELE, TINE

Distractor names (male):

ARNO, BJORN, CARLO, FILIP, HUGO, IVAN, JACOB, KURT, LARS, LUDO, MAURO,
OSCAR, OTMAR, RALF, SIMON, STAN, WALT

Distractor names (female):

AMBER, CINDY, DINA, ELINE, ELSA, EVIE, FIEN, FLOR, HANNE, HEDDA, HILDE,
JENTE, JULIA, LENA, MAUD, NAOMI, OLGA, PETRA, RITA, ROOS

T2 names (male):

BRAM, JELTE, JENS, NICK

T2 names (female):

ANKE, ANYA, ELIEN, EMMA, FIEN, FLEUR, HANNE, IRIS, LARA, LIEN, LIES, LUCIE,
MARIE, NINA, ROOS, SARA, TESSA, VERA

Appendix B

T2 Stimuli Experiment 2

T1 words:

ACCENT (accent)	GRENS (border)	RADIO (radio)
AGENDA (diary)	HENDEL (lever)	REKKEN (stretch)
BAAS (boss)	HERDER (shepherd)	RUIT (diamond)
DADEL (date)	IVOOR (ivory)	SCHROEF (screw)
DELEN (share)	KALM (calm)	SERVIES (service)
ENTRÉE (entrance)	LINIAAL (ruler)	STUDIO (studio)
EXACT (exact)	MARMER (marble)	SUBTIEL (subtle)
FATSOEN (decency)	NOTITIE (note)	TAK (branch)
FORNUIS (stove)	OVERLOOP (landing)	VIJVER (pond)
GARAGE (garage)	PLEIN (square)	ZOOL (sole)

T2 words referring to numbers:

DRIE (three), TIEN (ten), TWAALF (twelve), VIER (four), VIJF (five), ZEVEN (seven)

Random neutral T2 words:

BLAD (leaf), ETEN (eat), GELUID (sound), LUCHT (sky), TANTE (aunt), WATER (water)

Distractors:

AANBEVELINGSBRIEF (letter of recommendation)	ONONTCIJFERBAAR (undecipherable)
AANGELEGENHEID (matter)	ONTSLUIEREN (unveil)
AMBACHTSGEZEL (journeyman)	ONVERANTWOORDELIJK (irresponsible)
AMFITHEATER (amphitheater)	ORGANISATIE (organization)
ANTROPOLOGIE (anthropology)	PONTIFICEREN (pontificate)
AUTOBIOGRAFIE (autobiography)	POSTZEGELALBUM (stamp album)
BEATIFICATIE (beatification)	PROGNOSTICEREN (prognosticate)
BESCHRIJVING (description)	PROPORTIONEEL (proportional)
BINNENVAART (inland navigation)	PURIFICATIE (purification)
BUITENGEWOON (extraordinary)	RAADSELACHTIG (enigmatic)
CATALOGISERING (to catalogue)	RANGSCHIKKING (classification)
CHRYSANTEMUM (chrysanthemum)	RECHTVAARDIGING (justification)
CLASSIFICATIE (classification)	SCHEENBESCHERMER (shin guard)
COALITIETIEGERING (coalition government)	STAMBOEKDIER (pedigree animal)
CONGREGATIE (congregation)	STERRENWACHT (observatory)
CONTORTIONIST (contortionist)	TAFELTENNIS (table tennis)
DEMOGRAFIE (demography)	TRANSPORTATIE (transportation)
DESINTEGREREN (disintegrate)	TUSSENPERSOON (middleman)
DUBBELGANGER (doppelgänger)	UITKIJKTOREN (watchtower)
EGALISATIEFONDS (equalization fund)	UITMUNTEND (excellent)
EMPIRISCH (empirical)	VERDERFELIJK (pernicious)
ENCYCLOPEDIE (encyclopedia)	VERDUIDELIJKEN (clarify)
ETABLISSEMENT (establishment)	VERDWIJNING (disappearance)
EXTRAPOLEREN (extrapolate)	VERWARRING (confusion)
FEDERALISERING (federalization)	VOORAFSCHADUWEN (foreshadowing)
FOTOGRAFIE (photography)	WELOPGEVOED (well-mannered)
FRACTIEVERGADERING (meeting of a parliamentary party)	WINTERLANDSCHAP (winter landscape)
GELIJKTIJDIG (simultaneous)	ZAKENMAN (businessman)
GENERALISATIE (generalization)	ZEKERINGENPANEEL (fuse box)
GILDEBROEDERSCHAP (guild ship)	ZONSVERDUISTERING (solar eclipse)
GLOBALISATIE (globalization)	
HALLUCINATIE (hallucination)	
HERSCHIKKEN (rearrange)	
HYPOTHETISCH (hypothetical)	
INCUBATIETIJD (incubation period)	
INTELLECTUALISME (intellectualism)	
INTERNALISATIE (internalization)	
KALEIDOSCOOP (kaleidoscope)	
KIEZELSTRAND (shingle beach)	
KLASSEMENT (list of rankings)	
LOCALISATIE (localization)	
LOUTERING (chastening)	
MINISTERRAAD (council of ministers)	
MOTORFIETS (motorcycle)	
OBSERVATORIUM (observatory)	
OMSCHRIJVING (description)	
ONBEGRIJPELIJK (incomprehensible)	
ONBERISPELIJK (impeccable)	
ONDUBBELZINNIG (unambiguous)	